Type MS · Explain how beta-blockers and ACE inhibitors lower heart rate or pressure

Answer 1 of 6 · Question 7 part c · 3 marks

2024 JUN · Paper 02 · find part c in this mark scheme extract

7 (a)	Any four from: • less blood flow/blood flow slows/restricted; • to heart muscle/tissue/cells; • less oxygen/glucose; • less aerobic respiration; • heart muscle/tissue/cells die/angina/heart attack;		(4)
(b) (i)	• (women 29 ÷ 100 (x 33 500 000) 9 715 000; • (men 26 ÷ 100 (x 33 000 000) 8 580 000; • (9 715 000 – 8 580 000) 1 135 000/1.135 million/1.135×10^6;	ecf for mp3	(3)
(ii)	• calculate BMI; • mass÷(height)^2; • compare result to BMI chart/ obese if 30 or more;	Accept • waist: hip ratio; • waist÷hip • Compare result to hip:waist ratio chart/	(3)
(c)	• beta blockers bind/ attach to receptors (on cardiac muscle); • prevent neurotransmitters from binding/attaching (to receptors); • reduce heart rate/reduced blood pressure;	Accept beta blockers fits into receptors accept neurotransmitters do not fit into receptors	(3)

Total for Question 7 = 13 marks

Type MS · Explain how beta-blockers and ACE inhibitors lower heart rate or pressure

Answer 2 of 6 · Question 9 part a.i · 1 mark

2020 JAN · Paper 01 · find part a.i in this mark scheme extract

number				
9 (a)	(i)	gives competitors an unfair advantage/form of cheating (1)		1
	(ii)	Any four from • increases heart rate (1) • for greater blood flow/increases blood pressure (1) • glycogen to glucose (in muscles/liver)(1) • causes dilation of air passages/increased rate/depth of breathing for increased oxygen uptake (1) • redirects blood flow to muscles/brain/heart/lungs (1) • increases aerobic respiration (1)		4
(b)	(i)	• beta-blocker attaches to/binds with (adrenaline/norepinephrine/epinephrine) receptor sites (1) • binding of adrenaline/norepinephrine / epinephrine inhibited/prevented/reduced (1) • reduced response in cardiac muscle/tissue cells/named reduced response e.g. lower heart rate (1)	ignore ref to active site	3
	(ii)	Any two from: • slower heart beat (1) • reduced blood flow/oxygen/glucose to skeletal muscles (1) • slower rate of aerobic respiration (1) • less energy (for activity) (1)		2

Total for Question 9 = 10 marks

Type MS · Explain how beta-blockers and ACE inhibitors lower heart rate or pressure

Answer 3 of 6 · Question 9 part b.i · 3 marks

2020 JAN · Paper 01 · find part b.i in this mark scheme extract

Number					
9	(a)	(i)	gives competitors an unfair advantage/form of cheating (1)		1
		(ii)	Any four from • increases heart rate (1) • for greater blood flow/increases blood pressure (1) • glycogen to glucose (in muscles/liver)(1) • causes dilation of air passages/increased rate/depth of breathing for increased oxygen uptake (1) • redirects blood flow to muscles/brain/heart/lungs (1) • increases aerobic respiration (1)		4
	(b)	(i)	• beta-blocker attaches to/binds with (adrenaline/norepinephrine/epinephrine) receptor sites (1) • binding of adrenaline/norepinephrine / epinephrine inhibited/prevented/reduced (1) • reduced response in cardiac muscle/tissue cells/named reduced response e.g. lower heart rate (1)	ignore ref to active site	3
		(ii)	Any two from: • slower heart beat (1) • reduced blood flow/oxygen/glucose to skeletal muscles (1) • slower rate of aerobic respiration (1) • less energy (for activity) (1)		2

Total for Question 9 = 10 marks

Type MS · Explain how beta-blockers and ACE inhibitors lower heart rate or pressure

Answer 4 of 6 · Question 9 part b.ii · 2 marks

2020 JAN · Paper 01 · find part b.ii in this mark scheme extract

number				
9 (a)	(i)	gives competitors an unfair advantage/form of cheating (1)		1
	(ii)	Any four from • increases heart rate (1) • for greater blood flow/increases blood pressure (1) • glycogen to glucose (in muscles/liver)(1) • causes dilation of air passages/increased rate/depth of breathing for increased oxygen uptake (1) • redirects blood flow to muscles/brain/heart/lungs (1) • increases aerobic respiration (1)		4
(b)	(i)	• beta-blocker attaches to/binds with (adrenaline/norepinephrine/epinephrine) receptor sites (1) • binding of adrenaline/norepinephrine / epinephrine inhibited/prevented/reduced (1) • reduced response in cardiac muscle/tissue cells/named reduced response e.g. lower heart rate (1)	ignore ref to active site	3
	(ii)	Any two from: • slower heart beat (1) • reduced blood flow/oxygen/glucose to skeletal muscles (1) • slower rate of aerobic respiration (1) • less energy (for activity) (1)		2

Total for Question 9 = 10 marks

Type MS · Explain how beta-blockers and ACE inhibitors lower heart rate or pressure

Answer 5 of 6 · Question 9 part a · 4 marks

2022 JUN · Paper 01 · find part a in this mark scheme extract

9 (a)	• beta blockers; • block/bind to receptors; • which prevents the release of adrenaline/noradrenaline; • heart rate decreases; OR • ACE inhibitors; • reduce the activity of enzymes; • less angiotensin; • less narrowing of blood vessels;		4
(b)(i)	any two from • how much pressure blood is exerting; • against artery walls; • when ventricles contract/pump;		2
(ii)	• pressure in the arteries; • when the heart/ventricles at rest/relaxed/between beats;		2
			Total 8

Type MS · Explain how beta-blockers and ACE inhibitors lower heart rate or pressure

Answer 6 of 6 · Question 5 part a · 4 marks

2025 JUN · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

5 (a)	• ACE inhibitor prevents/inhibits action of enzyme; • prevents formation of angiotensin II; • less(angiotensin ii) bind with receptors; • Losartan blocks/binds to receptors; • angiotensin ii cannot bind to receptor;		4
(b)	• arteries/arterioles; • muscles in walls; • can contract/constrict; • narrow lumen; • smaller volume/space(for blood flow);		4
(c)	• in normal/healthy kidney protein not found in urine; • protein is a big molecule; • should not pass through the glomerulus/basement membrane; • damage to glomerulus/basement membrane;		2
(d)	• stronger/intense/eq; <u>contractions</u>; • (which causes) <u>muscle</u> to get thicker;	mp2 must be due to mp1	2
(e)	• detoxification occurs in the liver; • can't break down Losartan if impaired/accumulation /build-up in body/bloodstream/liver; • results in (further) liver damage/liver failure/liver diseae;		2 **Total =14**

Type MS · Explain how beta-blockers and ACE inhibitors lower heart rate or pressure

Answer 6 of 6 · Question 5 part a · 4 marks

2025 JUN · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Answer	Notes	Marks
	• A=rods/cones/retina;		